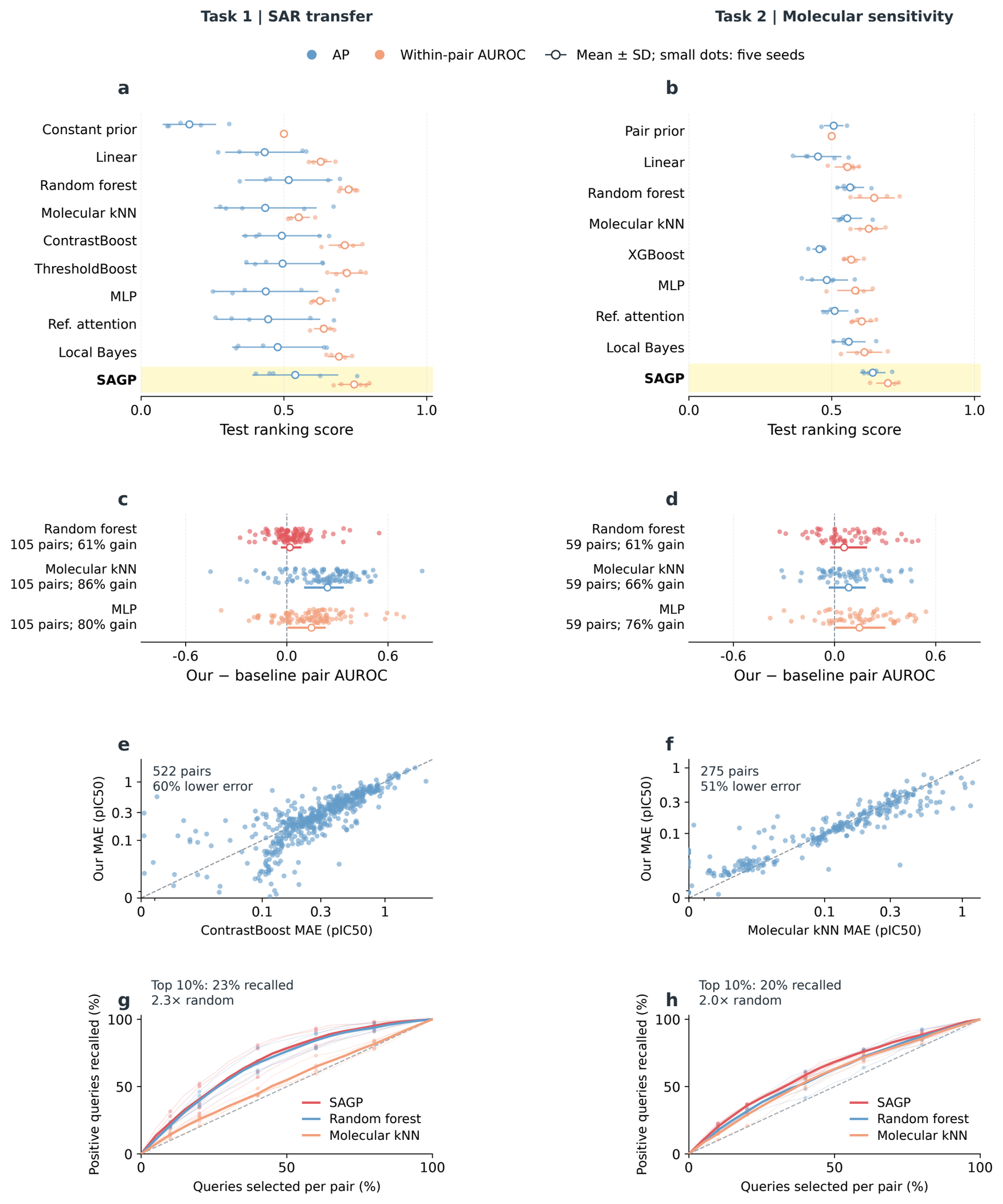

Task 1 | SAR transfer
Task 2 | Molecular sensitivity
AP
Within-pair AUROC
Mean ± SD; small dots: five seeds
a
Constant prior
Linear
Random forest
Molecular kNN
ContrastBoost
ThresholdBoost
MLP
Ref. attention
Local Bayes
SAGP
0.0
0.5
1.0
Test ranking score
b
Pair prior
Linear
Random forest
Molecular kNN
XGBoost
MLP
Ref. attention
Local Bayes
SAGP
0.0
0.5
1.0
Test ranking score
c
Random forest
105 pairs; 61% gain
Molecular kNN
105 pairs; 86% gain
MLP
105 pairs; 80% gain
-0.6
0.0
0.6
Our − baseline pair AUROC
d
Random forest
59 pairs; 61% gain
Molecular kNN
59 pairs; 66% gain
MLP
59 pairs; 76% gain
-0.6
0.0
0.6
Our − baseline pair AUROC
e
522 pairs
60% lower error
Our MAE (pIC50)
0
0.1
0.3
1
ContrastBoost MAE (pIC50)
f
275 pairs
51% lower error
Our MAE (pIC50)
0
0.1
0.3
1
Molecular kNN MAE (pIC50)
g
Top 10%: 23% recalled
2.3× random
Positive queries recalled (%)
0
50
100
SAGP
Random forest
Molecular kNN
Queries selected per pair (%)
h
Top 10%: 20% recalled
2.0× random
Positive queries recalled (%)
0
50
100
SAGP
Random forest
Molecular kNN
Queries selected per pair (%)